CSCI 2800 — Computer Architecture and Operating Systems (CAOS)
Lab 6 (document version 1.0)

- This lab is due by the **end of your lab session** on Wednesday, October 14, 2026
- This lab is to be completed **individually**; do **not** share your code with anyone else
- Do **not** use AI to assist you; instead, use posted examples, the Discussion Forum, trial-and-error, and your own intelligence
- You **must** show your code and your solutions to a TA, CA, or mentor to receive credit for each checkpoint
- Note that all of your C code **must** successfully compile via `gcc` with no warning messages when the `-Wall` (i.e., warn all) compiler option is used; also use `-Werror`, which treats all warnings as critical errors

1. **Checkpoints 1 and 2:** In support of Homework 2, your task in the first two checkpoints is to complete the code given in `lab06-checkpoint1.c`. Download and run this code; though it compiles with no warnings or errors and runs, it does not do anything useful yet. (In fact, it will likely crash due to memory not being properly initialized.)

   Specifically, implement the required functions described below. Be sure to use the given function prototypes exactly as shown. Comments in the given source file should help clarify what each function is to do. Also refer to `hw2-start.c` and your own work on Homework 2 thus far.

   Be sure that there are no memory leaks or buffer overflows or use of initialized memory. As per usual, do **not** use square bracket notation; instead, only use pointer arithmetic. If square brackets exist in the given code, you must remove them.

   Implement the following functions, most likely in the order shown:

   - **Checkpoint 1:**
     ```
     void upcase_word( char * word, int wordlen );
     int trie_index( char letter );
     ```
   - **Checkpoint 2:**
     ```
     void add_word( trie_t * trie, const char * word, int wordlen );
     int is_valid_word( const trie_t * trie, const char * word, int wordlen );
     void free_trie( trie_t * trie );
     ```

   Running your code should produce the following interactive spell-checking program, though the suggested word functionality is not required for Checkpoints 1 and 2:

```
bash$ ./a.out
Trie contains 11 valid words:
AA, AAH, AAHED, AAHING, AAHS, AAL, AALII, AALIIS, AALS, AARDVARK, CAOS
Enter word to check (CTRL-D to exit): CAOS
==> CAOS is valid
Enter word to check (CTRL-D to exit): CAORS
==> CAORS is not valid; did you mean CAOS?
Enter word to check (CTRL-D to exit): canos
==> CANOS is not valid                          <<<sorry, bad joke>>>
Enter word to check (CTRL-D to exit): aaLL
==> AALL is not valid; did you mean AALS?
Enter word to check (CTRL-D to exit): AARDVARKY
==> AARDVARKY is not valid; did you mean AARDVARK?
Enter word to check (CTRL-D to exit): AARDVA
==> AARDVA is not valid; did you mean AAH?
Enter word to check (CTRL-D to exit): <CTRL-D>
==> bye
bash$
```

   Note that if you have already implemented Homework 2 using a different approach, you are still required to meet the above requirements as stated.

2. **Checkpoint 3: At the beginning of your Wednesday lab section, this lab checkpoint will be available that contains further instructions.** As a hint, keep working on Homework 2 due this week.